

framework-comparison

Framework Comparison

Comprehensive comparison of all testing framework adapters in the pkg/userflow package. Use this guide to select the right adapter for your testing requirements.

Adapter Summary

Adapter	Interface	Platform	Tool	Source File
PlaywrightCLI-Adapter	BrowserAdapter	Web	Playwright + CDP	playwright_cli_adapter.go
SeleniumAdapter	BrowserAdapter	Web	Selenium WebDriver	selenium_adapter.go
CypressCLI-Adapter	BrowserAdapter	Web	Cypress CLI	cypress_adapter.go
PuppeteerAdapter	BrowserAdapter	Web	Puppeteer + Node.js	puppeteer_adapter.go
ADBCLIAdapter	MobileAdapter	Android	ADB	adb_cli_adapter.go
AppiumAdapter	MobileAdapter	Android, iOS	Appium 2.0	appium_adapter.go
Maestro-CLIAdapter	MobileAdapter	Android, iOS	Maestro CLI	maestro_adapter.go
EspressoAdapter	MobileAdapter	Android	Gradle + ADB	espresso_adapter.go
	BuildAdapter	Android (JVM)	Gradle	robolectric_adapter.go

Adapter	Interface	Platform	Tool	Source File
Ro-bolectricAdapter				
GRPCCLI-Adapter	GRPCAdapter	API	grpcurl	adapter_grpc.go
GorillaWebSocketAdapter	WebSocketFlowAdapter	API	gorilla/websocket	adapter_websocket_flow.go
HTTPAPI-Adapter	APIAdapter	API	pkg/httpclient	http_api_adapter.go
TauriCLIAdapter	DesktopAdapter	Desktop	Tauri Web-Driver	tauri_cli_adapter.go
GoCLIAdapter	BuildAdapter	Build	go build/test	go_cli_adapter.go
GradleCLI-Adapter	BuildAdapter	Build	Gradle	gradle_cli_adapter.go
NPMCLIAdapter	BuildAdapter	Build	npm/npx	npm_cli_adapter.go
CargoCLI-Adapter	BuildAdapter	Build	Cargo	cargo_cli_adapter.go
ProcessCLI-Adapter	ProcessAdapter	System	subprocess	process_cli_adapter.go

Feature Matrix: Browser Adapters

Feature	Playwright	Selenium	Cypress	Puppeteer
Chrome	Yes	Yes	Yes	Yes
Firefox	Yes	Yes	Yes	Experimental
Safari/WebKit	Yes (WebKit)	Yes	No	No
Edge	Yes (Chromium)	Yes	Yes	No
Headless mode	Yes	Yes	Yes	Yes
Network intercept	Via CDP (not CLI)	No (use proxy)	No (CLI)	No (CLI)
Screen-shots	Yes	Yes (base64)	Yes (file)	Yes (base64)

Feature	Playwright	Selenium	Cypress	Puppeteer
JavaScript eval	Yes	Yes	Yes (base64)	Yes (base64)
Session model	CDP Web-Socket	HTTP session	Per-spec process	WS endpoint
Speed	Fast	Medium	Slow	Medium
Container exec	podman exec	HTTP server	npx (local)	podman exec fallback
Setup complexity	Low	Medium (Grid)	Medium (npm)	Low
State persistence	Per-CDP connection	Per-session	None	Per-browser

Feature Matrix: Mobile Adapters

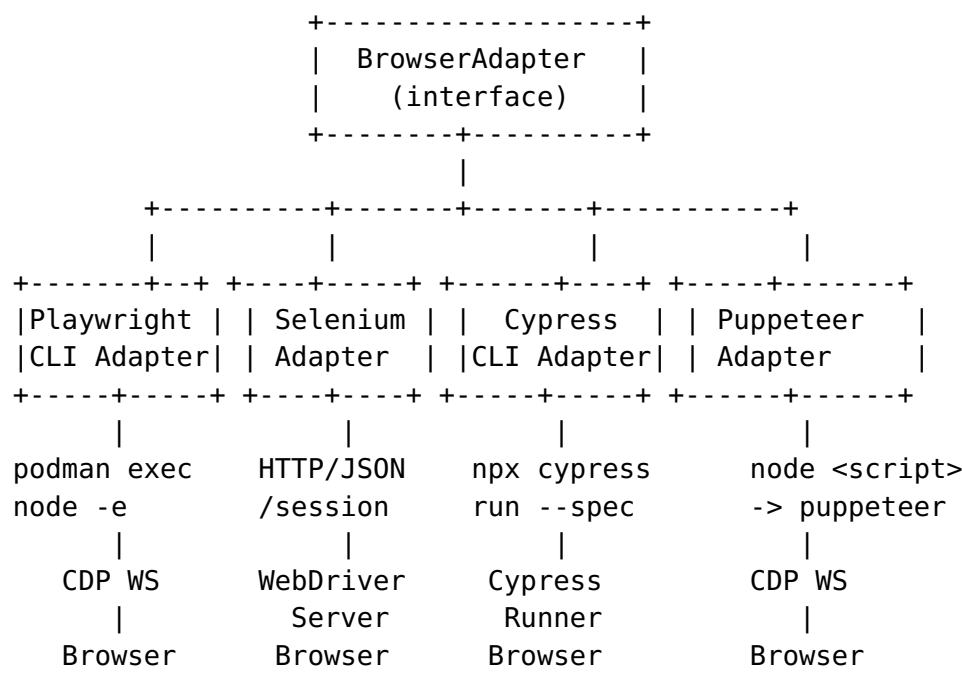
Feature	ADB CLI	Appium	Maestro	Espresso
Android	Yes	Yes	Yes	Yes
iOS	No	Yes	Yes	No
Device required	Yes	Yes	Yes	Yes
Emulator support	Yes	Yes	Yes	Yes
No device (JVM)	No	No	No	No
Instrumented tests	No	Via shell	No	Yes (native)
Element selectors	No	W3C Web-Driver	Text/accessibility	Espresso matchers
Screen-shots	Yes (exec-out)	Yes (base64)	Yes (file)	Yes (exec-out)
Cross-platform	Android only	Android + iOS	Android + iOS	Android only
Setup complexity	Low (ADB only)	High (server)	Low (CLI)	Medium (Gradle)
Protocol	ADB shell	W3C Web-Driver	CLI/YAML	ADB + Gradle
Speed	Fast	Medium	Medium	Medium

Feature Matrix: API and Protocol Adapters

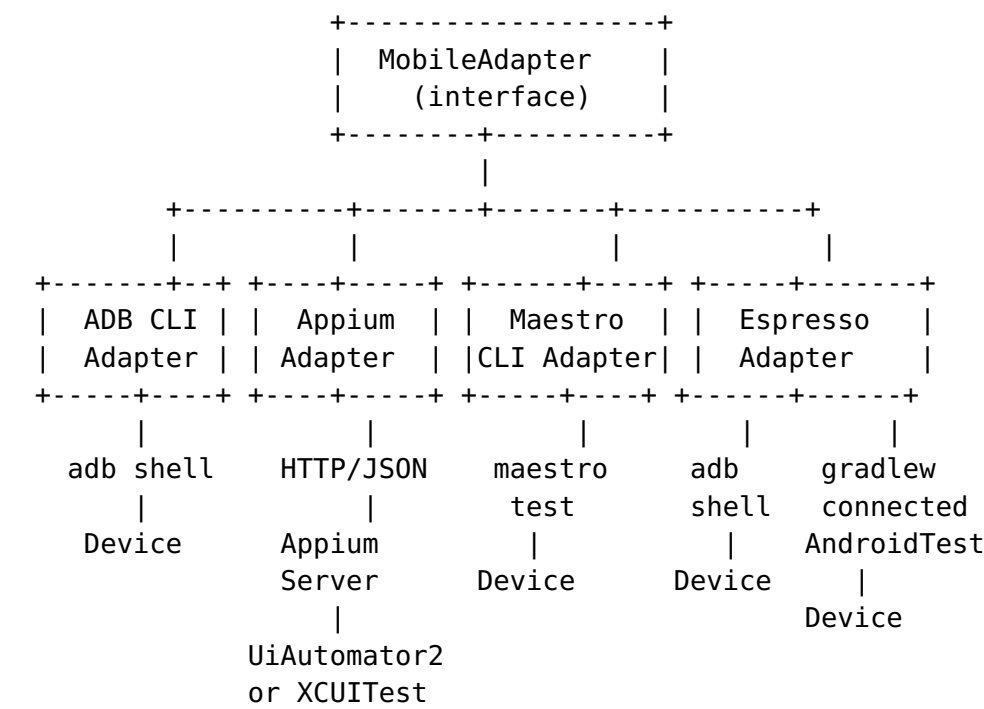
Feature	HTTP API	gRPC CLI	WebSocket
Protocol	HTTP/HTTPS	gRPC (HTTP/2)	WebSocket
Request type	REST JSON	Protobuf (JSON wire)	Binary/Text frames
Streaming	SSE (read only)	Server streaming	Bidirectional
Auth support	JWT, API key	Headers, TLS certs	Headers on connect
Connection	Per-request	Per-command	Persistent
Thread-safe	Yes (http.Client)	N/A (CLI)	Yes (mutexes)
Health check	HTTP status	grpc.health.v1	Connection check
Tool required	None (stdlib)	grpcurl	gorilla/websocket

Architecture Diagrams

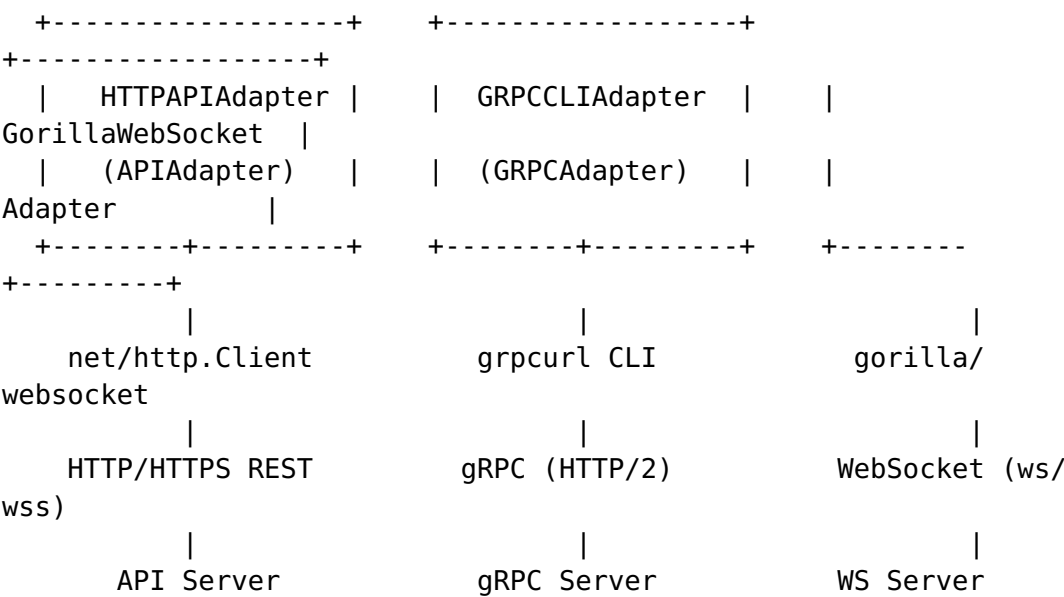
Browser Adapter Architecture



Mobile Adapter Architecture



Protocol Adapter Architecture



When to Use Each Framework

Browser Testing

Scenario	Recommended Adapter
	Playwright

Scenario	Recommended Adapter
General web testing with container infrastructure	
Multi-browser testing (including Safari)	Selenium
Project already uses Cypress	Cypress
Simple scripting with Node.js	Puppeteer
CI/CD with minimal dependencies	Selenium (standalone Docker)

Mobile Testing

Scenario	Recommended Adapter
Android-only, fast device interaction	ADB CLI
Cross-platform (Android + iOS)	Appium
Quick UI flow verification	Maestro
Instrumented test execution	Espresso
JVM unit tests, no device	Robolectric
Full E2E with element queries	Appium

Protocol Testing

Scenario	Recommended Adapter
REST API testing	HTTP API
gRPC service testing	gRPC CLI
Real-time WebSocket flows	WebSocket
SSE event streams	HTTP API

Cost and Licensing

Framework	License	Cost
Playwright	Apache 2.0	Free
Selenium	Apache 2.0	Free
Cypress	MIT	Free (open source), paid dashboard
Puppeteer	Apache 2.0	Free
Appium	Apache 2.0	Free
Maestro	Apache 2.0	Free (CLI), paid cloud
Espresso	Apache 2.0	Free (part of AndroidX)

Framework	License	Cost
Robolectric	MIT	Free
grpcurl	MIT	Free
gorilla/websocket	BSD-2-Clause	Free

All adapters in this package use open-source tools with permissive licenses. No adapter requires a paid service for local execution.

CI/CD Considerations

Adapter	Docker Available	Emulator Needed	Typical CI Time
Playwright	Yes (mcr.microsoft.com/playwright)	No	Fast (~seconds/test)
Selenium	Yes (selenium/standalone-chrome)	No	Medium
Cypress	Yes (cypress/included)	No	Slow (~seconds/spec)
Puppeteer	Yes (custom Node.js)	No	Medium
Appium	Yes (appium/appium)	Yes (Android)	Slow
Maestro	Partial	Yes	Medium
Espresso	Partial	Yes	Slow
Robolectric	Yes (any JDK image)	No	Fast
gRPC CLI	Yes (fullstorydev/grpcurl)	No	Fast
WebSocket	Yes (any Go image)	No	Fast

Documentation Index

Adapter	Documentation
Playwright	browser-adapter.md
Selenium	selenium-adapter.md
Cypress	cypress-adapter.md
Puppeteer	puppeteer-adapter.md
ADB CLI	mobile-adapter.md

Adapter	Documentation
Appium	<u>appium-adapter.md</u>
Maestro	<u>maestro-adapter.md</u>
Espresso	<u>espresso-adapter.md</u>
Robolectric	<u>robolectric-adapter.md</u>
gRPC CLI	<u>grpc-adapter.md</u>
WebSocket	<u>websocket-adapter.md</u>
HTTP API	<u>api-adapter.md</u>
Desktop (Tauri)	<u>desktop-adapter.md</u>
Build (Go/npm/Gradle/Cargo)	<u>build-adapter.md</u>
Process	<u>process-adapter.md</u>